

IMPORTANT STUDENT INSTRUCTIONS

Ensure your **Student Number** and **date of birth** are **entered** in the appropriate fields on **EVERY** sheet you use, including pink continuation sheets.

Use **BLACK INK** only. DO NOT USE BLUE INK OR PENCIL.

MUST BE COMPLETED

Student number (7 numbers)

2 8 3 9 7 6 8
N N N N N N N

Date of Birth (DD/MM/YYYY)

0 9 / 0 3 / 2 0 0 4
D D M M Y Y Y Y

COMPULSORY Q1.

Do not mark any other boxes.

☒ Q1

☐ DO NOT USE

☐ DO NOT USE

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☐ DO NOT USE

You may use both sides but only answer **ONE** question per sheet

Q1

(a)

(i) Eq 1: $\frac{mgl \sin \theta}{2} + I \frac{d^2 \theta}{dt^2} + c \frac{d\theta}{dt} = \tau$

since $\theta \ll 1$, $\sin \theta \approx \theta$, then:

$\Rightarrow \ddot{\theta} + \frac{c}{I} \dot{\theta} + \frac{mgl}{2I} \theta = \frac{\tau}{I}$

Laplace transform:

$\Rightarrow \Theta(s) \cdot s^2 + \frac{c}{I} \Theta(s) \cdot s + \frac{mgl}{2I} \Theta(s) = \frac{1}{2} T(s)$

$\Rightarrow G(s) = \frac{\Theta(s)}{T(s)} = \frac{\frac{1}{2}}{s^2 + \frac{c}{I}s + \frac{mgl}{2I}} = \frac{\frac{3}{4\pi^2}}{s^2 + \frac{3c}{4\pi^2}s + \frac{3g}{2l}}$

(ii) By using $m = 5 \text{ kg}$, $l = 0.4 \text{ m}$, $g = 9.81 \frac{\text{m}}{\text{s}^2}$, $c = 2 \text{ Nm/rad}$, $I = \frac{\pi^2}{3}$

$G(s) = \frac{3.75}{s^2 + 7.5s + 36.7875} = \frac{3.75}{s^2 + 2\xi\omega_n s + \omega_n^2}$

$\Rightarrow \begin{cases} 2\xi\omega_n = 7.5 \\ \omega_n^2 = 36.7875 \end{cases} \Rightarrow \begin{cases} \omega_n = 6.065 \text{ rad/s} \\ \xi = 0.618 \end{cases}$

(b) By using observer canonical form:

$A_o = \begin{bmatrix} -7.5 & 1 \\ -36.7875 & 0 \end{bmatrix}$ $B_o = \begin{bmatrix} 0 \\ 3.75 \end{bmatrix}$

$C_o = [0 \ 1]$ $D_o = 0$

Therefore:

$\begin{bmatrix} \ddot{\theta} \\ \dot{\theta} \end{bmatrix} = A_o \begin{bmatrix} \dot{\theta} \\ \theta \end{bmatrix} + B_o \frac{u}{\cancel{\pi^2}}$, $y = C_o \begin{bmatrix} \dot{\theta} \\ \theta \end{bmatrix} + D_o u$

$\Rightarrow \begin{bmatrix} \ddot{\theta} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -7.5 & 1 \\ -36.7875 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \theta \end{bmatrix} + \begin{bmatrix} 0 \\ 3.75 \end{bmatrix} u$

and $y = [0 \ 1] \begin{bmatrix} \dot{\theta} \\ \theta \end{bmatrix} + 0 u$

Q1 Continued

You may use both sides but only answer **ONE** question per sheet

(c). Two differences :

① The states with initial condition can only be expressed in state-space model but not in the transfer function.

② The steady-state can be calculated through transfer function easily.

(d).

For a system that is completely controllable, we can get its transfer function through the controller canonical form, which is :

$$\text{for } G(s) = \frac{b_1}{s^2 + a_1 s + a_2} = \frac{100}{s^2 - 5.5s + 220}$$

$$A_0 = \begin{bmatrix} -a_1 & -a_2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 5.5 & -220 \\ 1 & 0 \end{bmatrix}$$

$$B_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad C_0 = \begin{bmatrix} 0 & 100 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \ddot{\theta} \\ \dot{\theta} \end{bmatrix} = A_0 \begin{bmatrix} \dot{\theta} \\ \theta \end{bmatrix} + B_0 u$$

$$\Rightarrow \begin{bmatrix} \ddot{\theta} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 5.5 & -220 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \theta \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 100 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \theta \end{bmatrix} + 0 u$$

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